Auteur: Siebe Mees
Studierichting: Informatica
Oefeningenreeks 4A nr. 52.4

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\begin{aligned}
&\int \sin^4 x \cos^2 x dx \\
&= \int \left(\frac{1-\cos 2x}{2}\right)^2 \left(\frac{1+\cos 2x}{2}\right) dx \\
&= \frac{1}{8} \int \left(1 - 2\cos 2x + \cos^2 2x\right)\left(1+\cos 2x\right) dx \\
&= \frac{1}{8} \int \left(1 - \cos 2x - \cos^2 2x + \cos^3 2x\right) dx \\
&= \frac{1}{8} \int \left(1 - \cos 2x - \frac{1+\cos 4x}{2} + \cos 2x \frac{1+\cos 4x}{2}\right) dx \\
&= \frac{1}{8} \int \left(1 - \cos 2x - \frac{1}{2} - \frac{1}{2}\cos 4x + \frac{1}{2}\cos 2x + \frac{1}{2}\cos 2x \cos 4x\right) dx \\
&= \frac{1}{8} \int \left(\frac{1}{2} - \frac{1}{2}\cos 2x - \frac{1}{2}\cos 4x + \frac{1}{2}\cos 2x \cos 4x\right) dx \\
&= \frac{1}{8} \int \left(\frac{1}{2} - \frac{1}{2}\cos 2x - \frac{1}{2}\cos 4x + \frac{1}{4}\left(\cos 6x + \cos 2x\right)\right) dx \\
&= \frac{1}{8} \int \left(\frac{1}{2} - \frac{1}{4}\cos 2x - \frac{1}{2}\cos 4x + \frac{1}{4}\cos 6x\right) dx \\
&= \int \left(\frac{1}{16} - \frac{1}{32}\cos 2x - \frac{1}{16}\cos 4x + \frac{1}{32}\cos 6x\right) dx \\
&= \frac{x}{16} - \frac{1}{32}\frac{\sin 2x}{2} - \frac{1}{16}\frac{\sin 4x}{4} + \frac{1}{32}\frac{\sin 6x}{6} + C \\
&= \frac{x}{16} - \frac{\sin 2x}{64} - \frac{\sin 4x}{64} + \frac{\sin 6x}{192} + C
\end{aligned}
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References